Start here

Mathematics for AI and Machine Learning | KLAi Professional Learning

Welcome. This course teaches the mathematics under machine learning and AI, one idea at a time: a picture, a small calculation by hand, the same idea on a computer, and where it shows up in tools such as ChatGPT, Gemini and Claude. You need high school algebra and nothing else.

What you get

- **Slides as PDF.** Read them again after class. They make sense without the speaker.
- **A notebook for every session.** For learners who code. Open it in Google Colab (free) with a Google account.
- **A workbook for every session.** For everyone else, and for coders who like spreadsheets. It works in Excel and in Google Sheets.
- **A Check sheet or check cells.** They tell you which of your answers are right. You can try as often as you like.
- **A formula sheet template.** You fill it in as the course goes, in your own words.

No exams, no marks, and what you keep

- There are no exams, marks, grades or pass marks in this course. The check cells and the Check sheet say OK or NOT YET so that you can see what works. They are for you, and nothing is recorded.
- The recall and exit-check questions are for you to test yourself. Nobody marks them.
- In Session 18 you build a course project from the five builds, on your own personal data, and you keep it. A short project checklist helps you and a partner give each other feedback.
- When you complete the course project you receive a certificate of completion from KLAi Professional Learning.

How a session runs

Minutes	What happens
10	Recall, then a real question from AI
35	Concept: a picture, then one example worked by hand
10	Break
45	Lab in Python, Google Sheets or Excel
15	Where this appears in AI, and an exit check
5	Save your work

Using the notebook

- Upload the notebook to Colab (File, Upload notebook). Run the cells from the top with Shift and Enter.
- Each step is PREDICT, RUN, READ, CHANGE. Write your guess before you run. Type your answer in the line that starts with a greater-than sign.
- Cells with "your code here" are yours to fill. The check cells say OK or NOT YET, and finish with how many checks pass.
- Before you finish, choose Runtime, then Restart session and run all.

Using the workbook

- Open the .xlsx in Excel. In Google Sheets choose File, Import, Upload, then Replace spreadsheet.
- Gold cells are yours to fill. Everything else is given. The sheet called Check tells you which answers are right.
- Use the formulas shown on the Start here sheet. No add-ins, no macros.

Using AI tools while you learn

You may ask an assistant such as ChatGPT, Gemini or Claude to explain an idea. Check what it says against your own numbers from the lab. Never paste private or confidential data into an assistant. The course project is yours to build and keep.